

Transcriptome assembly and differential expression analysis on *Callosobruchus maculatus*

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Introduction

- Background -

- In *Callosobruchus maculatus*, a member of Coleoptera, Y-linked variation plays a significant role in male body size
- Males with two additional TOR (target of rapamycin) copies in the Y chromosome are smaller than those with one TOR copy

- Aims of this project-

This project focused on:

- Processing RNA samples from different *C. maculatus* individuals
- Building a transcriptome
- Observing different expression patterns among groups



Samples

- 82 individuals were sampled:
 - 26 females, 56 males
 - 39 from one TOR copy lineage*
 - 41 from three TOR copy lineage*
 - Each sample was performed in three lanes, forward and reverse
 - The lineages were inbred to prevent heterozygosity.
-

=> **Illumina RNA reads**



Preprocessing

- PIPELINE -

Illumina RNA reads



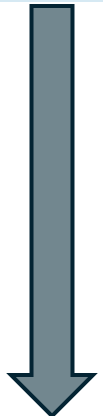
Initial
Quality Analysis:
FastQC



Adapter Trimming:
fastp

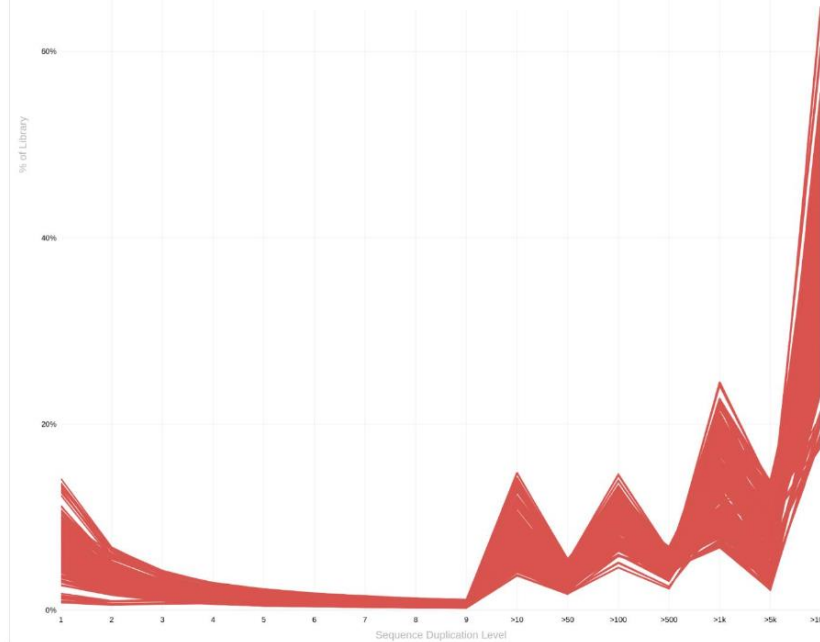


Sorting out rRNA:
SortMeRNA

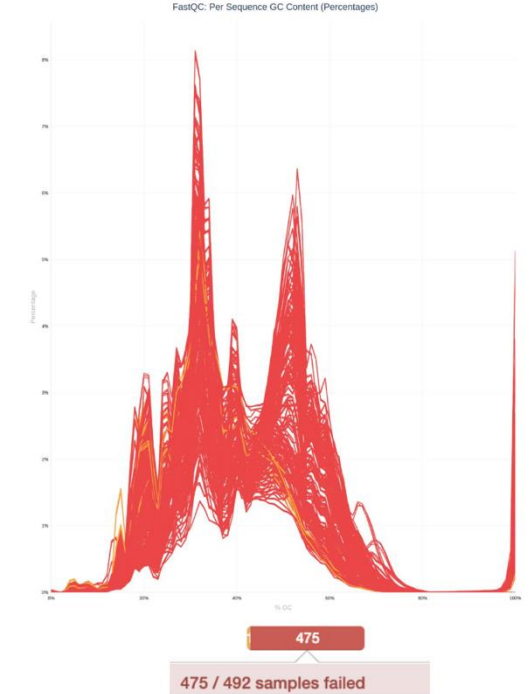


- Understanding the initial state of the data

Sequence Duplication Levels



GC content



Illumina RNA reads



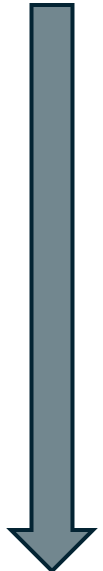
Initial
Quality Analysis:
FastQC



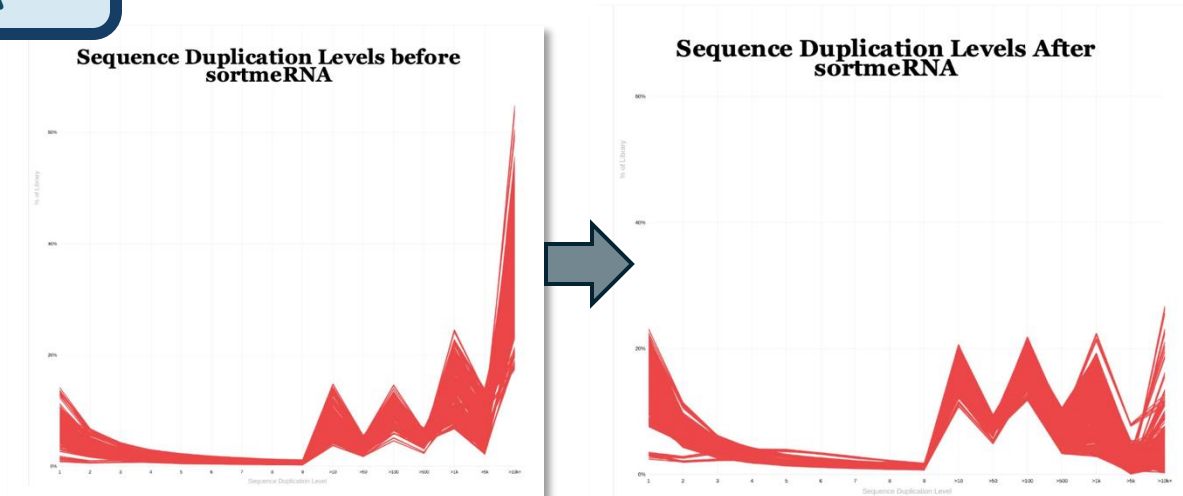
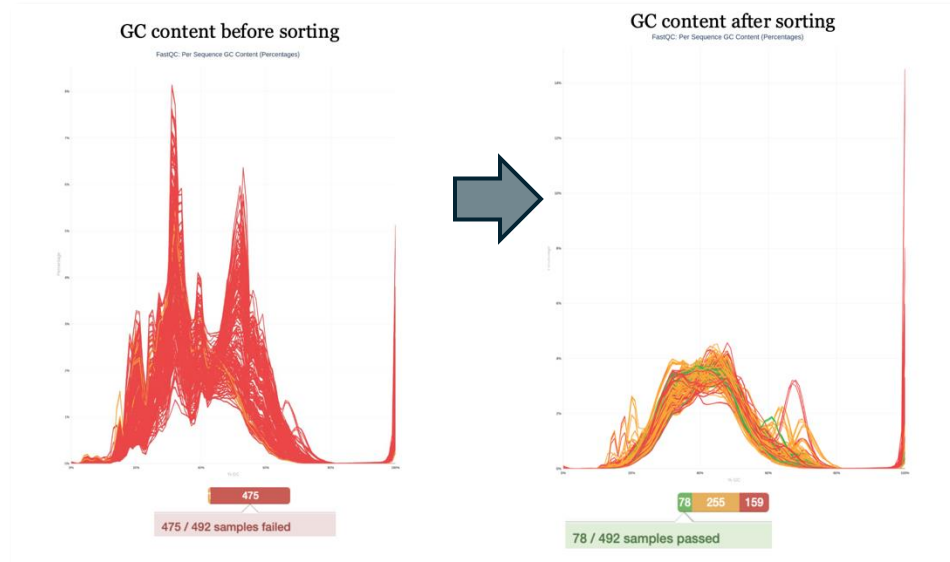
Adapter Trimming:
fastp



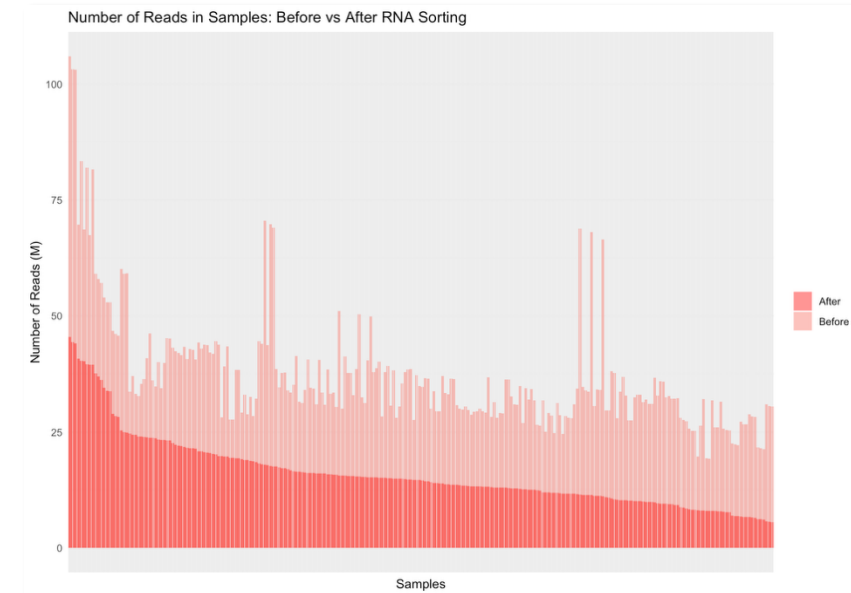
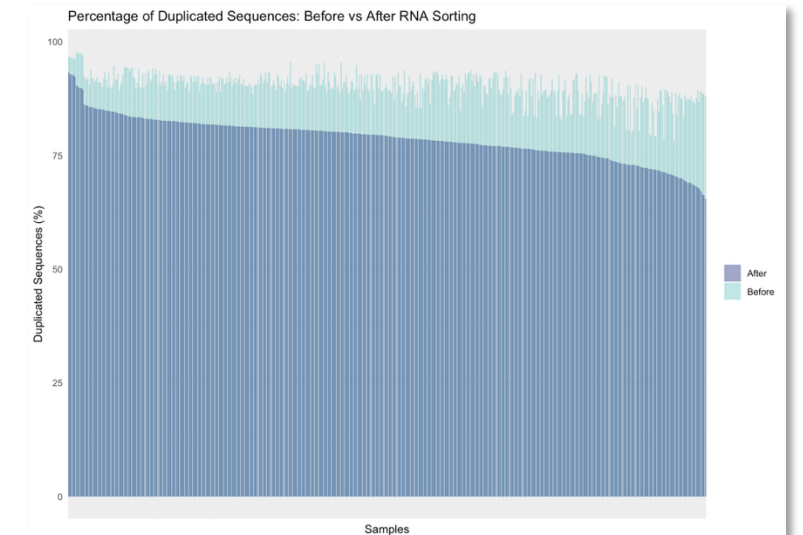
Sorting out rRNA:
SortMeRNA



- Results of sorting out rRNA:



Preprocessing

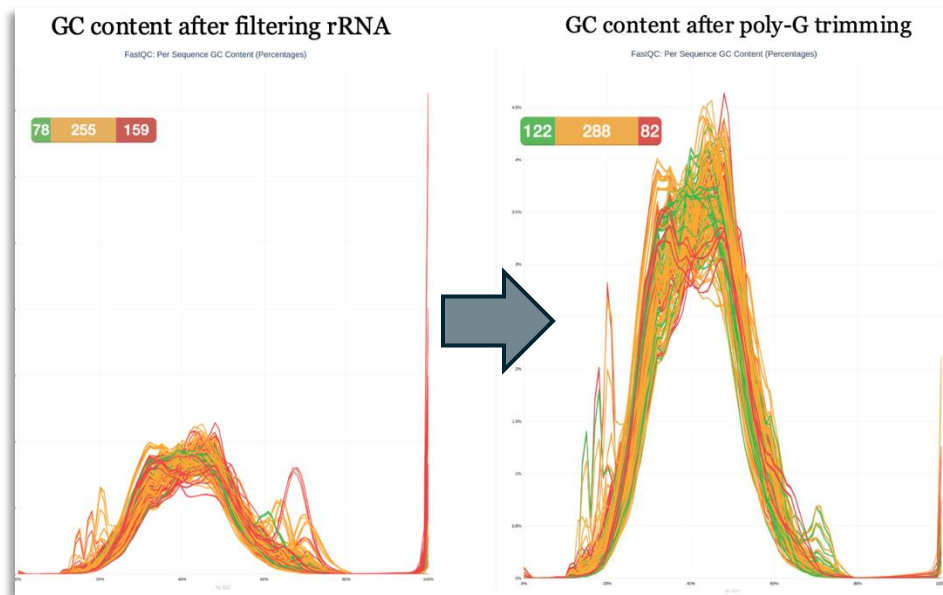


Sorting out rRNA:
SortMeRNA

Trimming out
polyG tails:
fastp

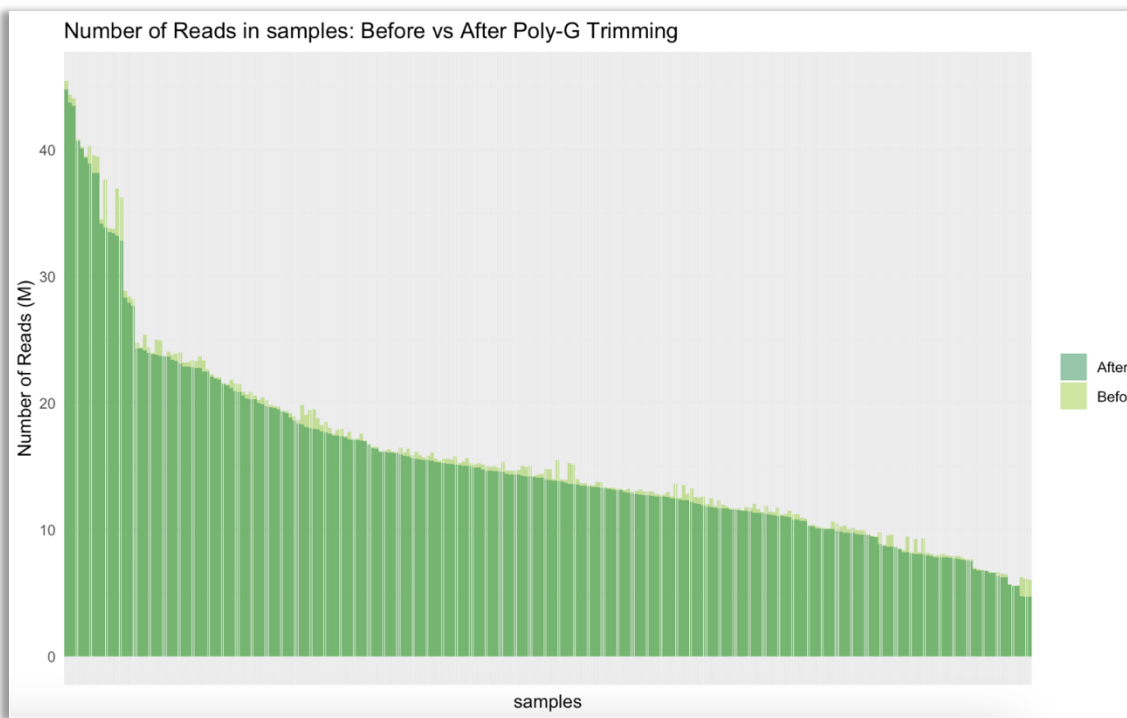
Collapsing the
read lanes

Reads deduplication:
seqkit

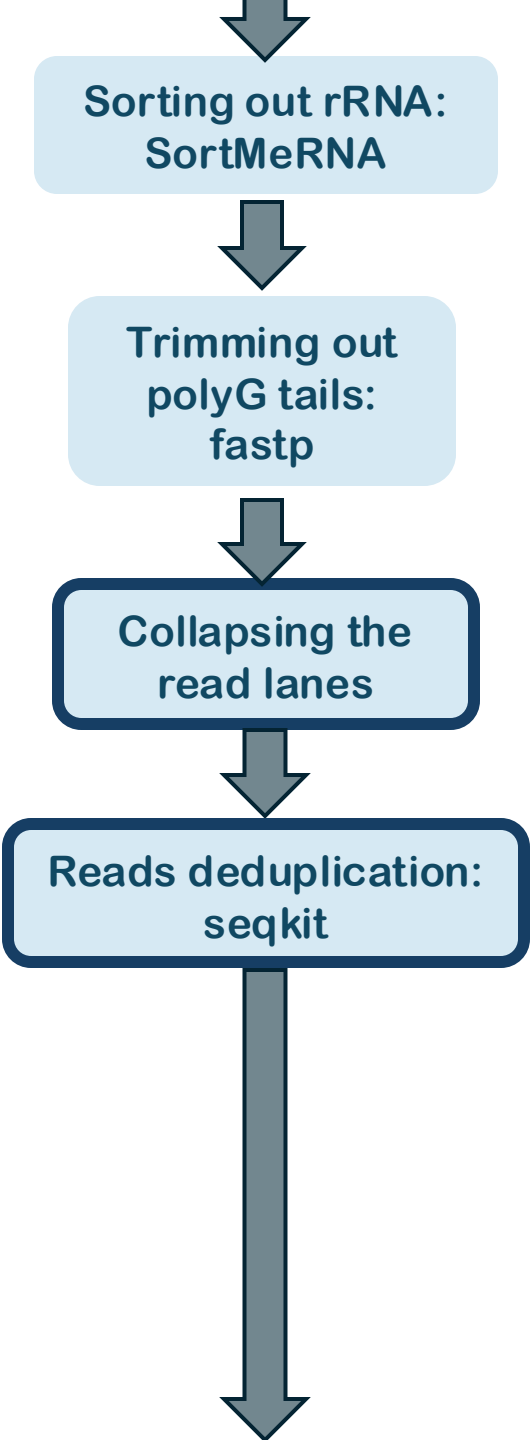


Preprocessing

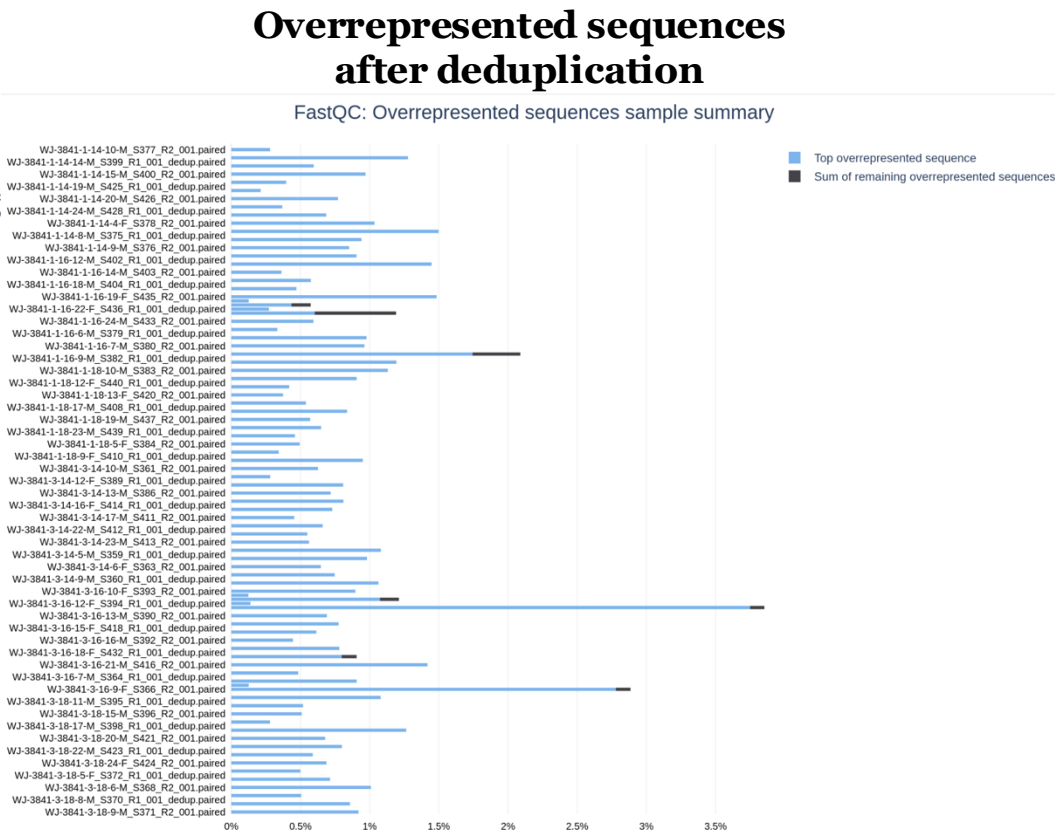
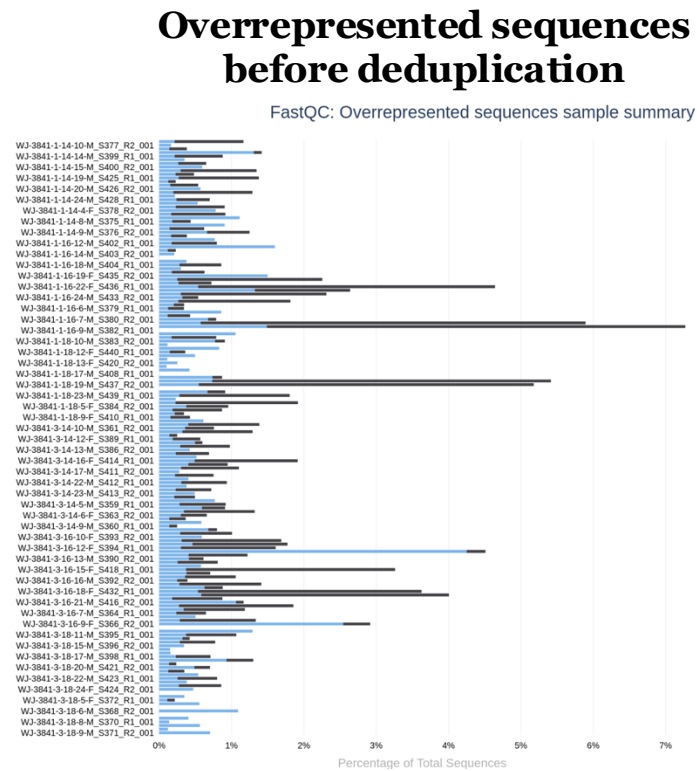
- At this point, 1.57% of all reads were GGGGGG repeat sequences. This was linked to polyG tails, an artifact caused by the two-channel Illumina sequencing system.



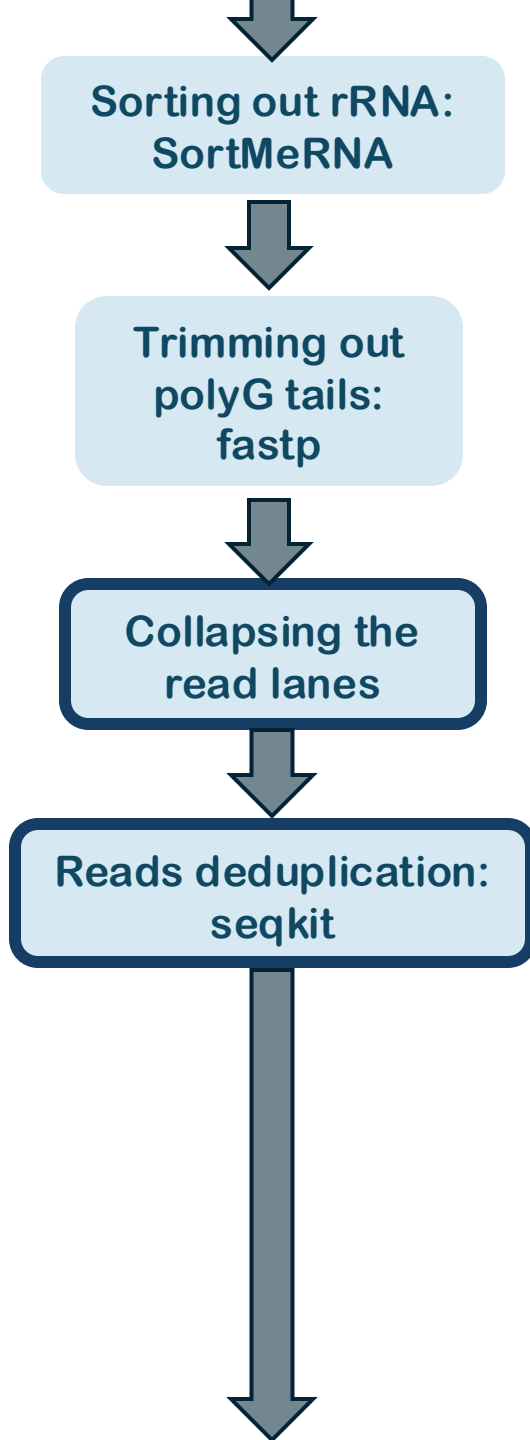
Preprocessing



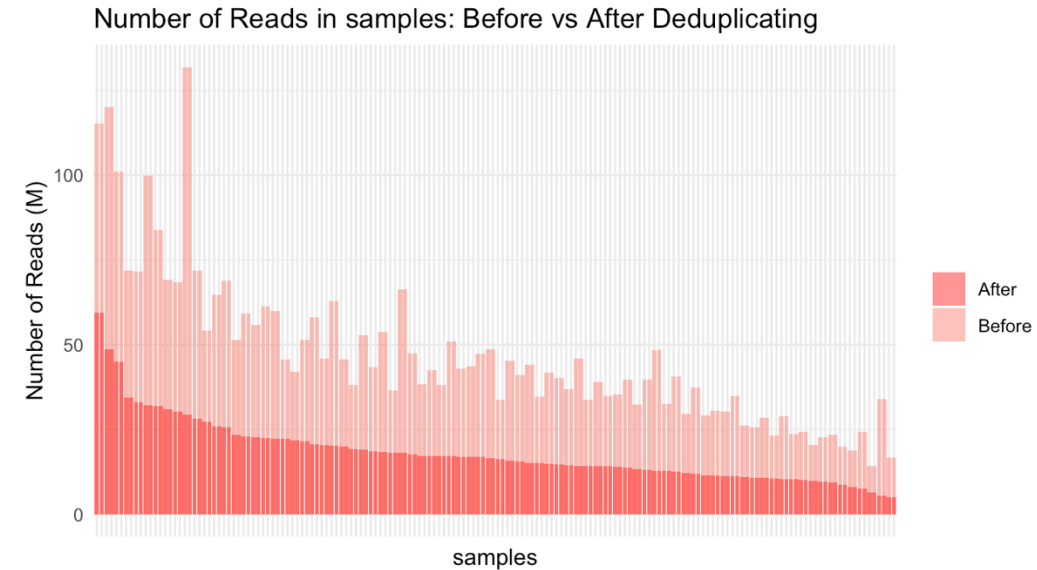
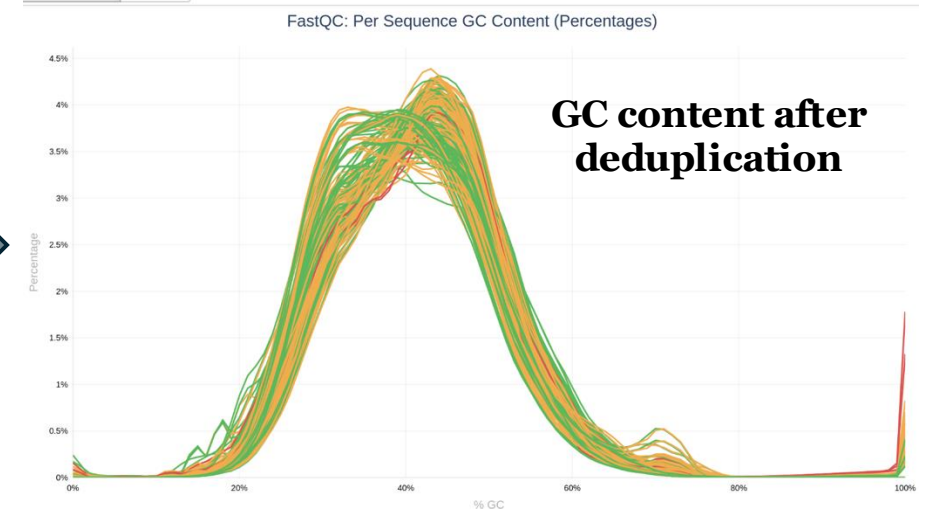
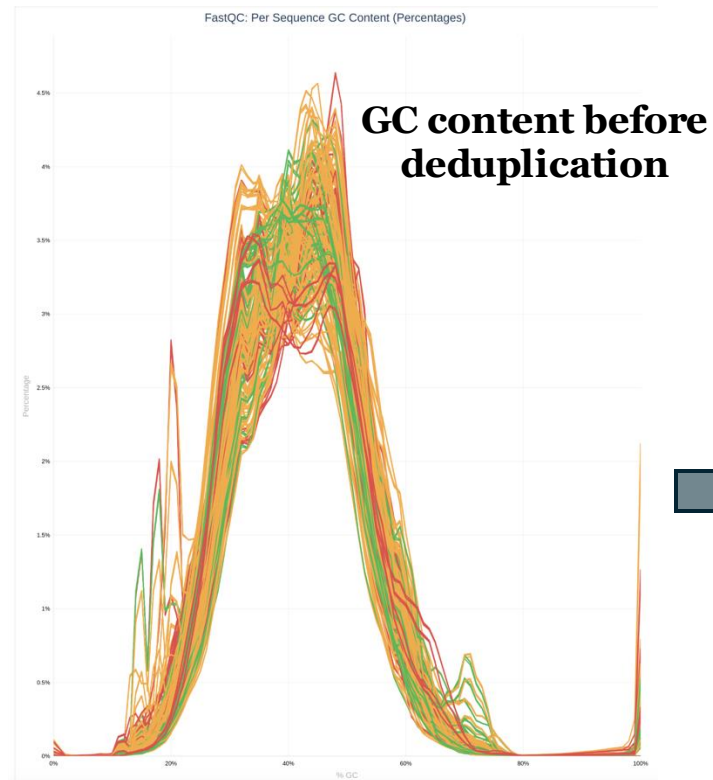
- At this point, a high number of duplications or overrepresented sequences were identified. However, they were linked to PCR duplications rather than true duplications, since exact duplicates are more likely to be PCR duplicates.



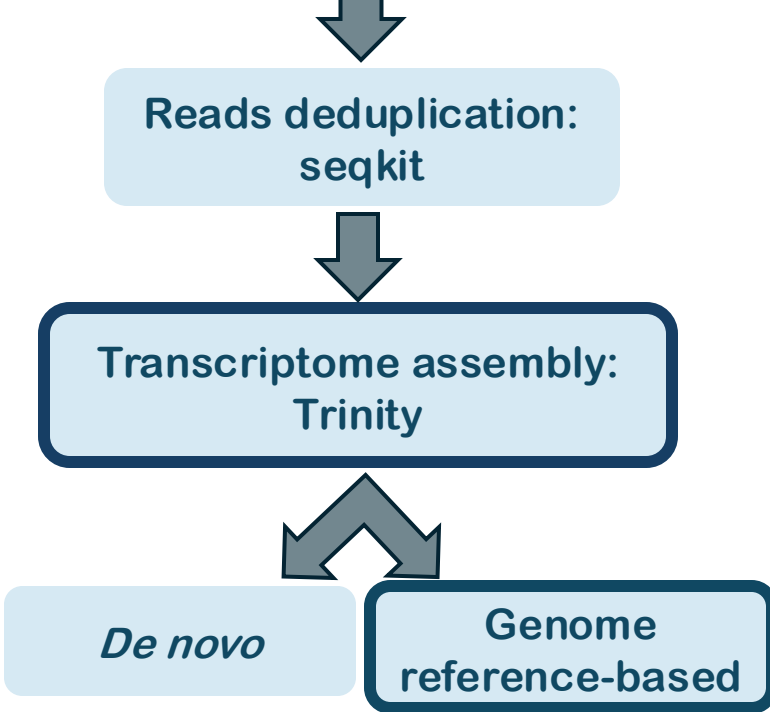
Preprocessing



- Multiple indicators showed that the deduplication was a good choice for the overall read quality, including GC content.



Transcriptome assembly



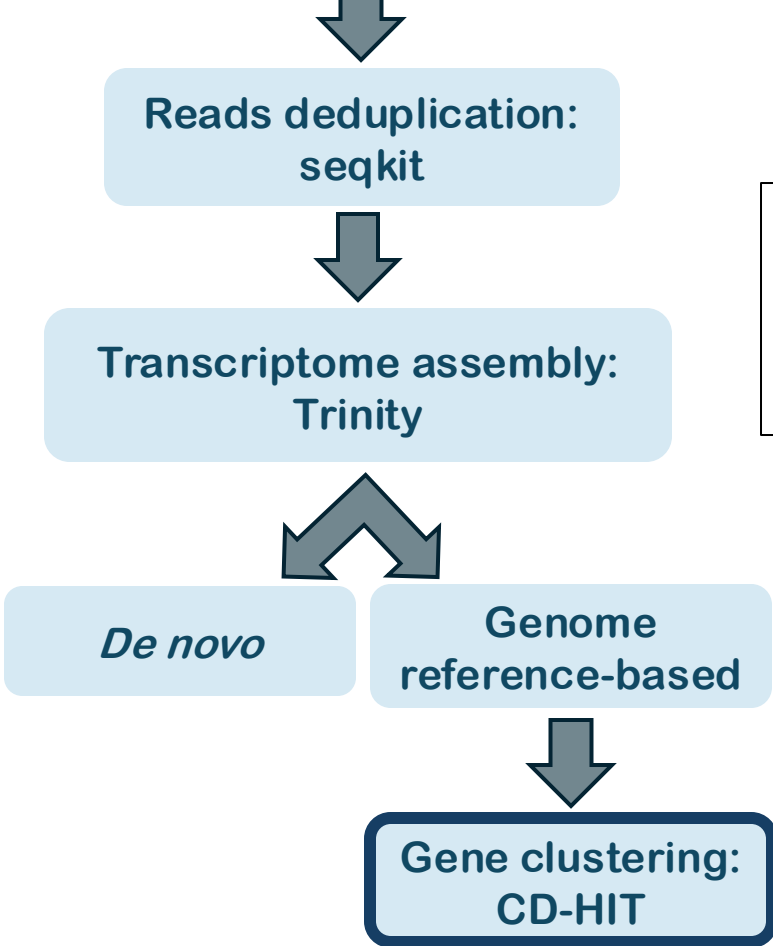
	Metric	De novo	Reference-based
Trinity statistics	Total Trinity 'genes'	173,983	262,864
	Total Trinity transcripts	294,829	344,749
	Percent GC	38.24%	37.50%
Mapping statistics	Reads alignment rate	98.10%	96.43%
BUSCO	Complete BUSCOs (C)	2014 (94.8%)	2034 (95.8%)
	└ Single-copy (S)	597 (28.1%)	550 (25.9%)
	└ Duplicated (D)	1417 (66.7%)	1484 (69.9%)
	Fragmented BUSCOs (F)	50 (2.4%)	34 (1.6%)
	Missing BUSCOs (M)	60 (2.8%)	56 (2.6%)

Too many genes, too many duplicates!

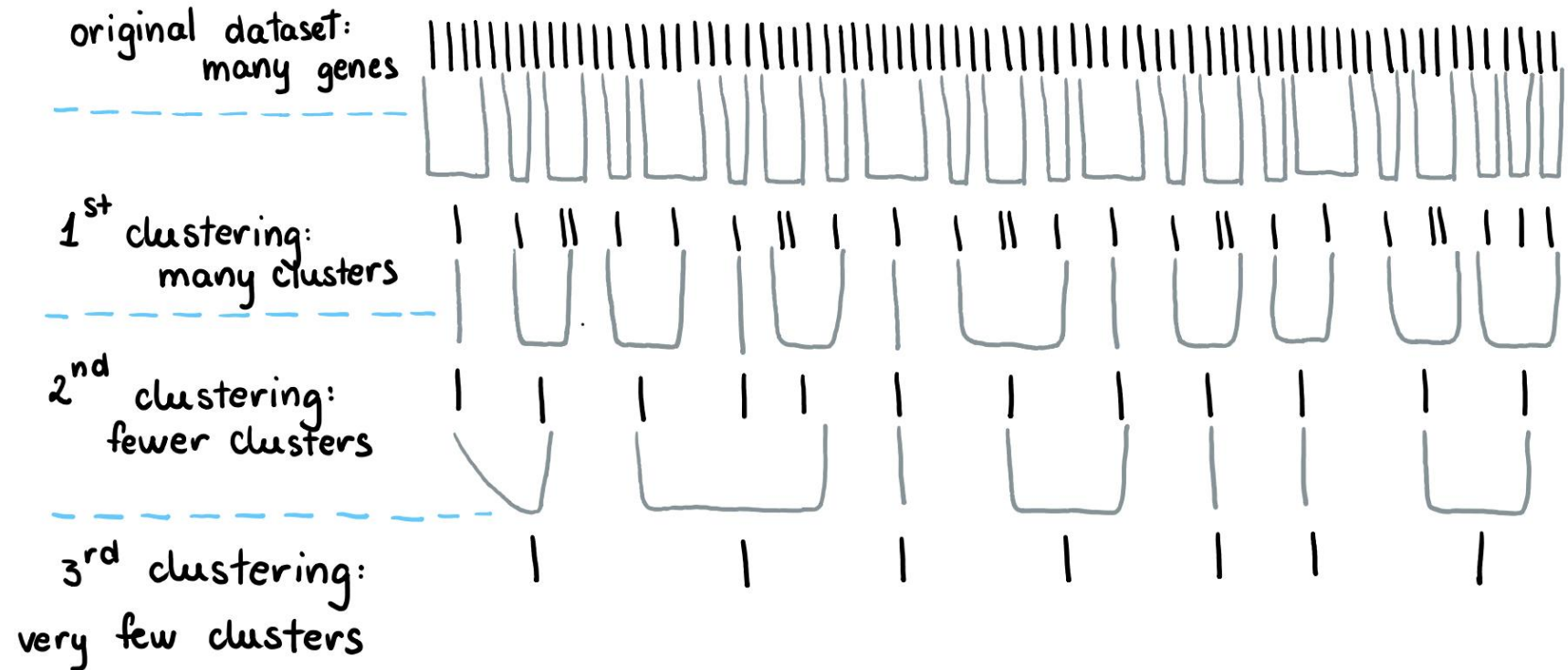
- Good mapping rate and BUSCO complete matches were found in both *de novo* and genome-based approaches.
- We went further with the reference-based transcriptome since it identified more Trinity genes

Transcriptome assembly

- To reduce the number of predicted genes, I designed a **hierarchical clustering algorithm** using CD-HIT. The same dataset would go through a clustering step iteratively, loosening the identity threshold each time.



Hierarchical clustering architecture:



Transcriptome assembly

Reads deduplication:
seqkit

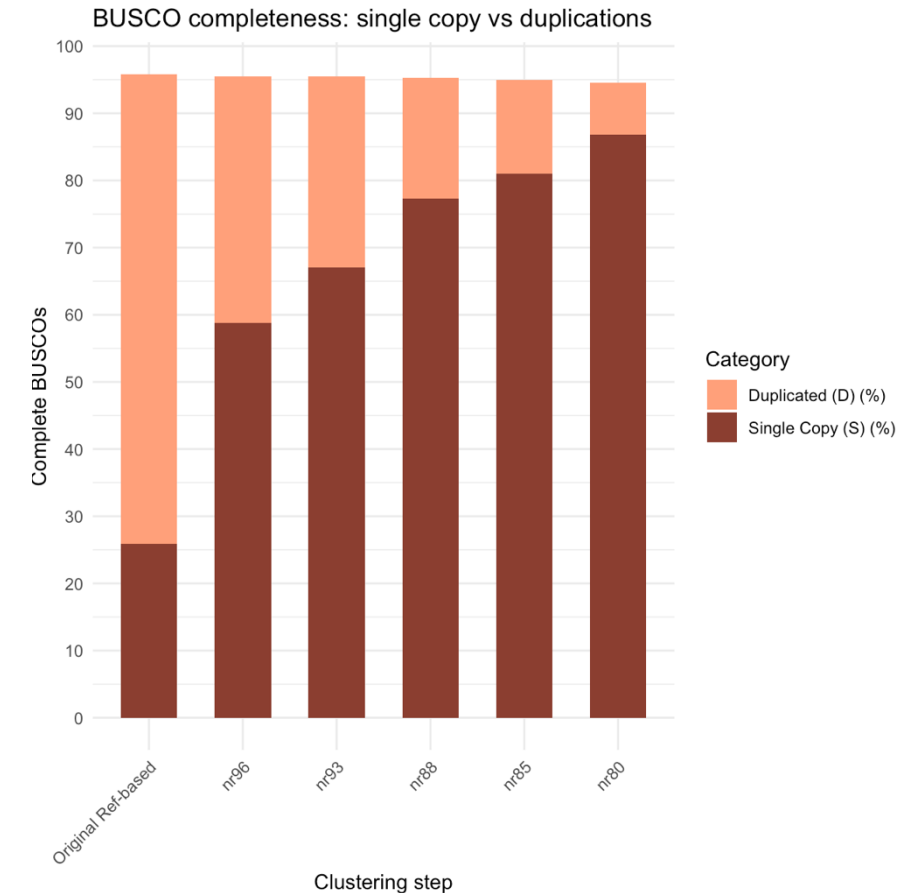
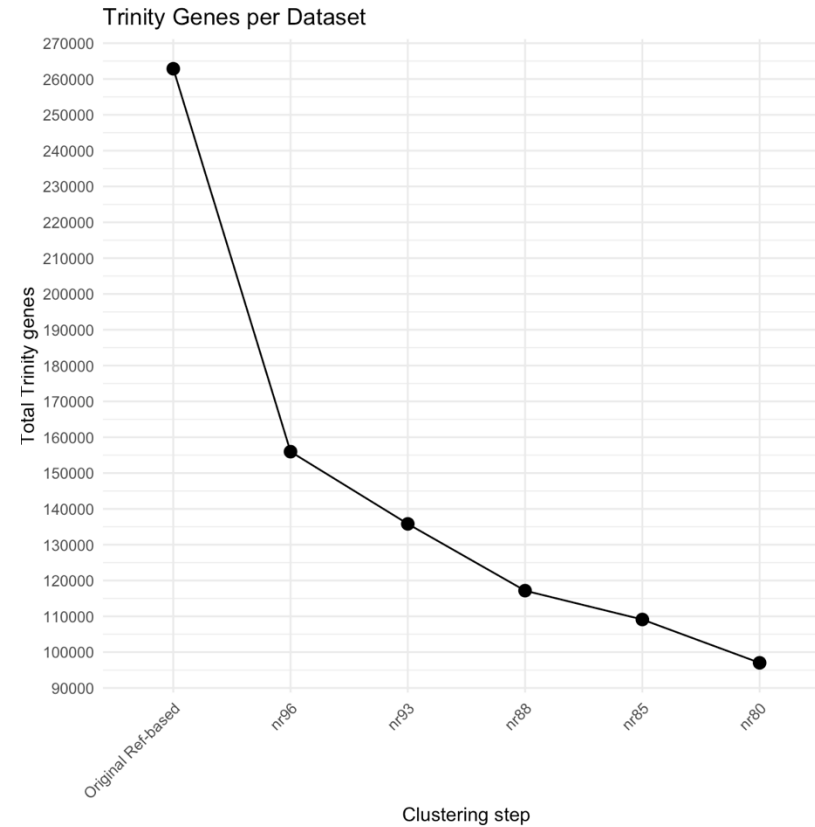
Transcriptome assembly:
Trinity

De novo

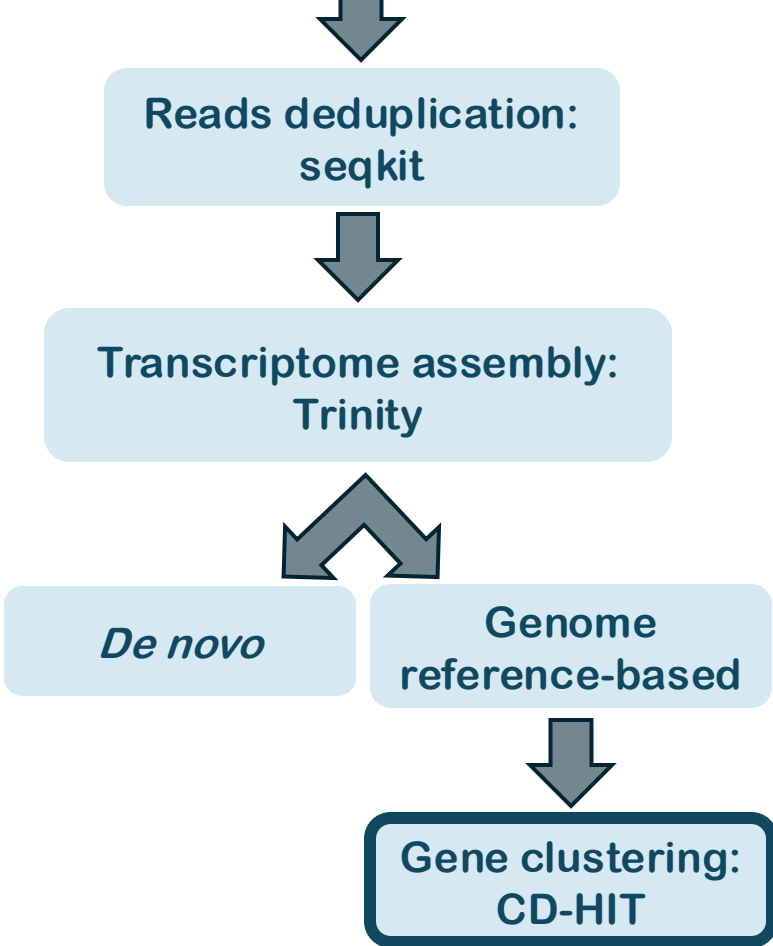
Genome
reference-based

Gene clustering:
CD-HIT

- After checking the literature on Cmac transcriptome assemblies, we decided to stop at identity threshold of 95% to keep some variability in the data.



Transcriptome assembly



RESEARCH ARTICLE

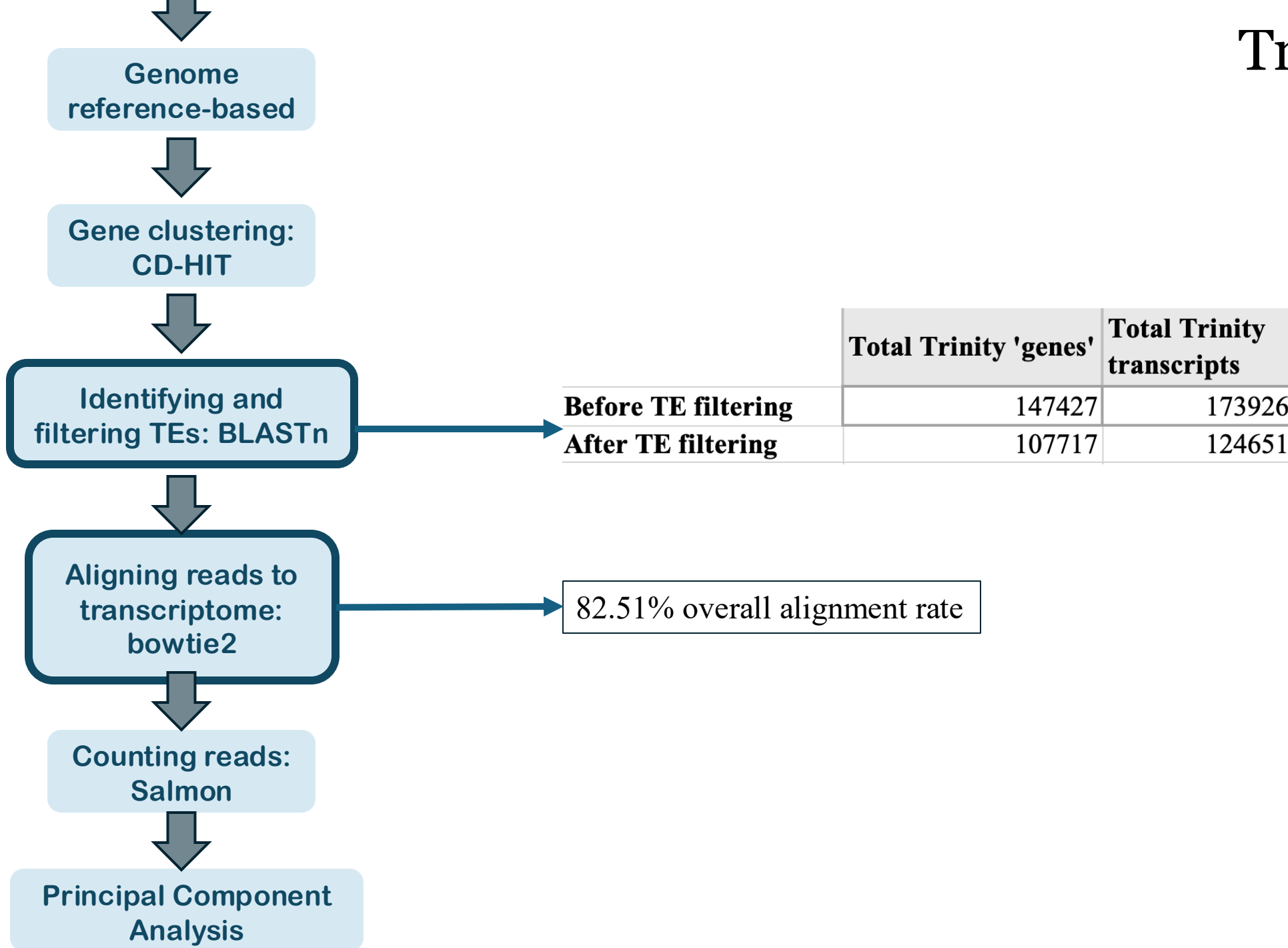
The *De Novo* Transcriptome and Its Functional Annotation in the Seed Beetle *Callosobruchus maculatus*

Ahmed Sayadi, Elina Immonen, Helen Bayram, Göran Arnqvist 

Published: July 21, 2016 • <https://doi.org/10.1371/journal.pone.0158565>

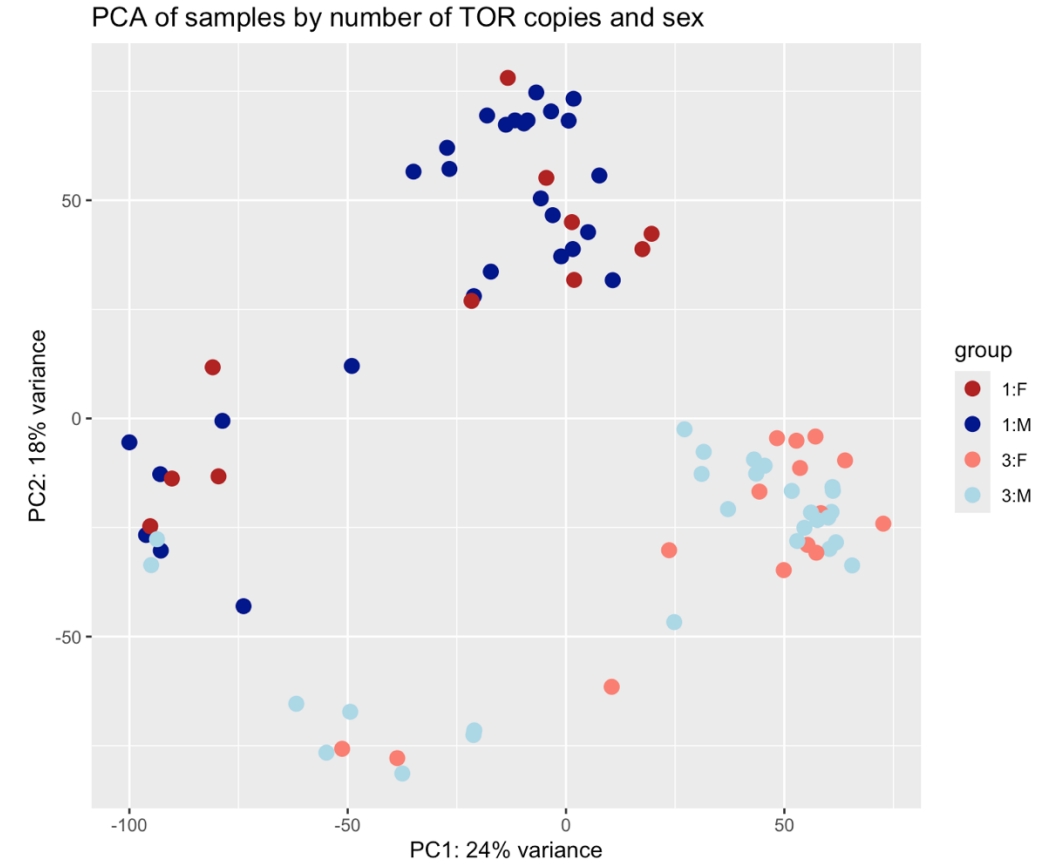
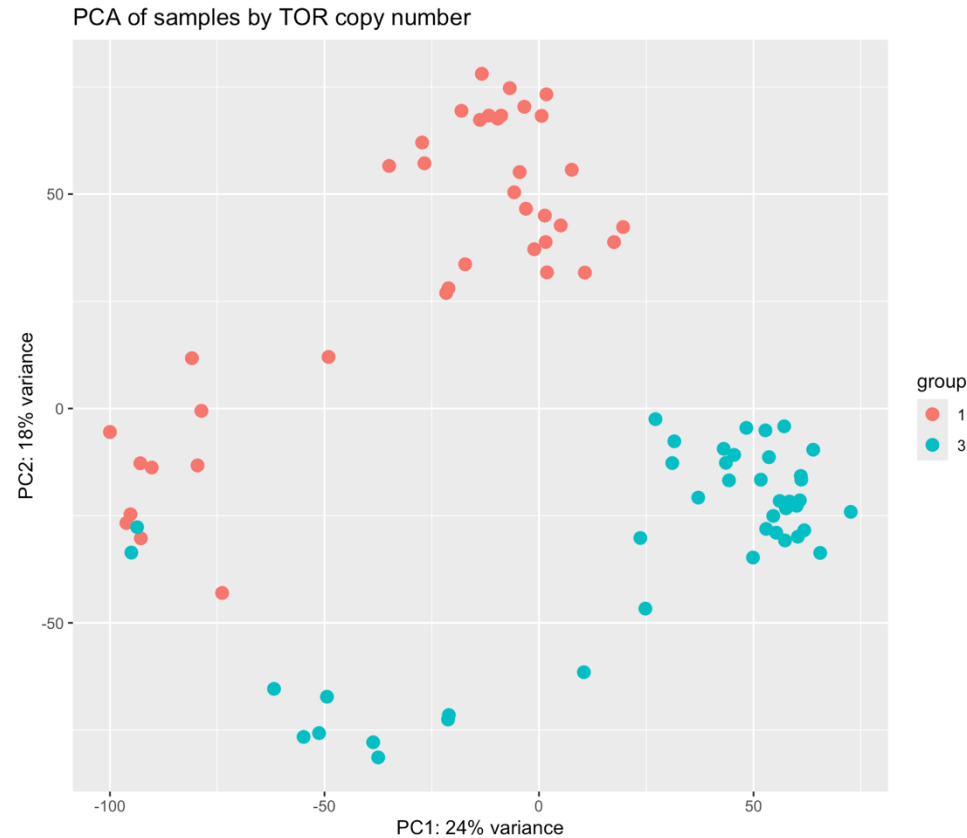
	Our transcriptome	Atricle transcriptome
Transcripts	173926	218192
Genes	147427	145883
BUSCO completeness	95.6%	90%
BUSCO duplicates	33.1%	50%

Transcriptome processing



Principal Component Analysis

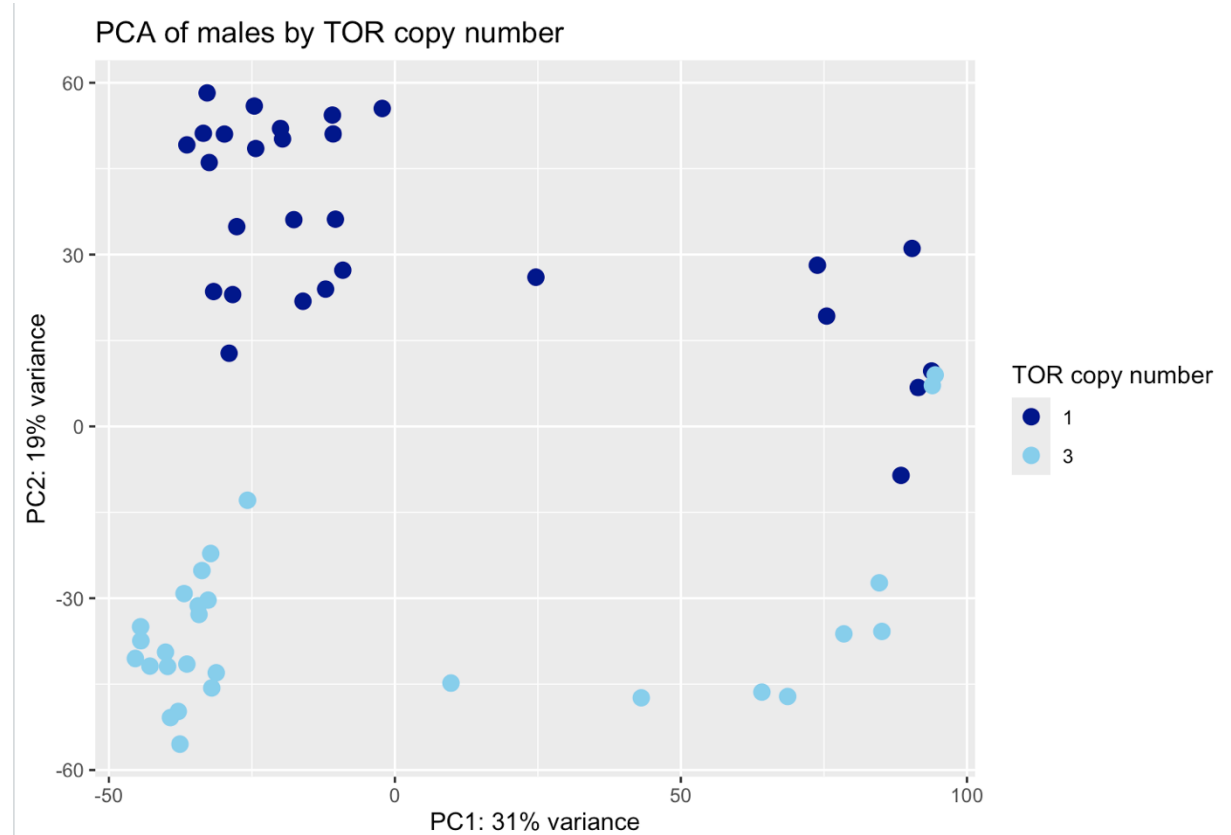
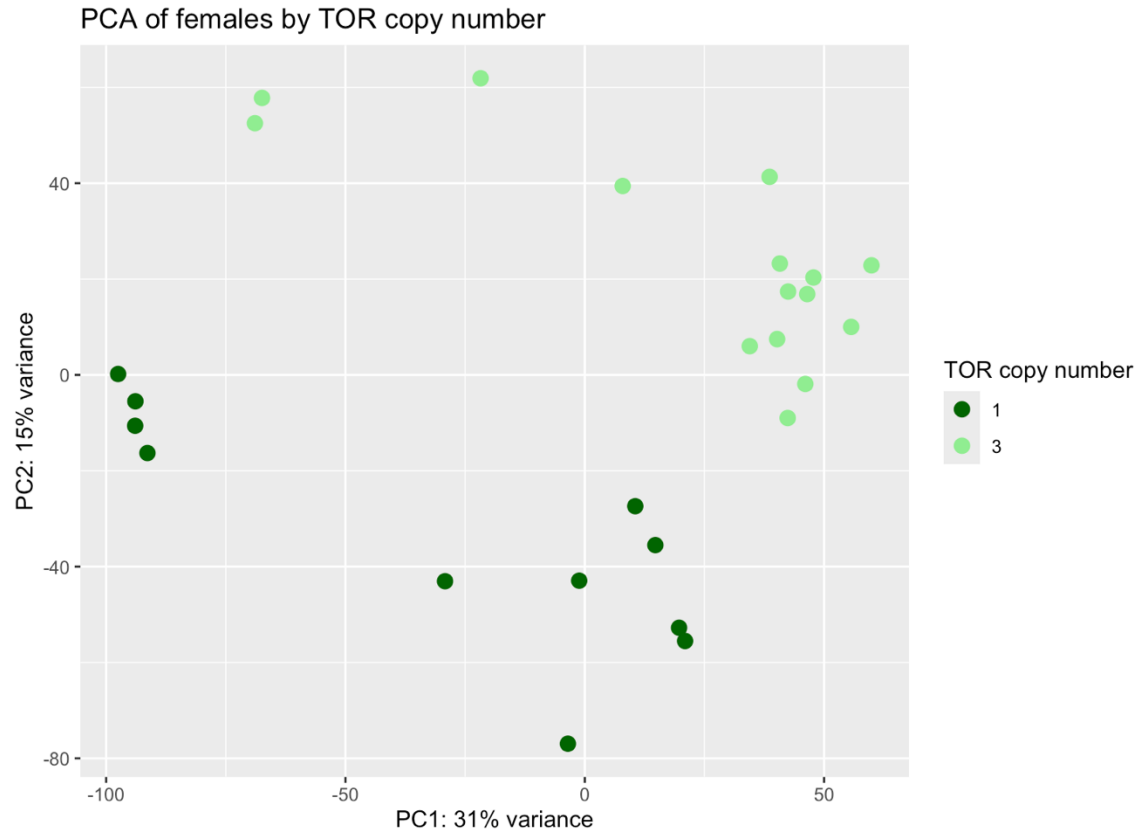
Differential expression analysis



- TOR copies number creates the clearest clustering. This is captured by a combination of both PCs.
- The fact that also females are separated could mean that there are differences in expression due to other factors, not necessarily the TOR copies

Principal Component Analysis

Differential expression analysis

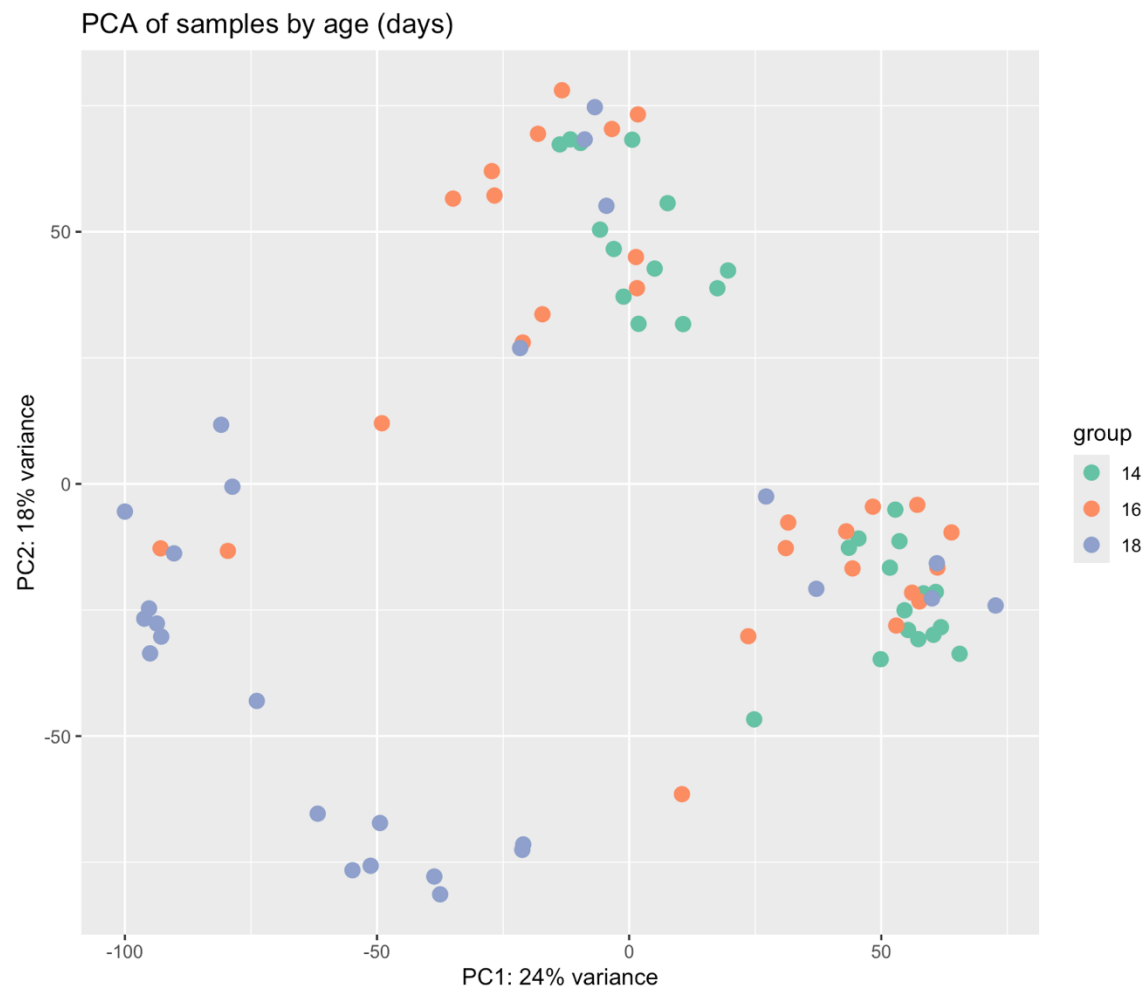


- The clustering by TOR copy number is slightly more significant in males than in females



Principal Component Analysis

Differential expression analysis

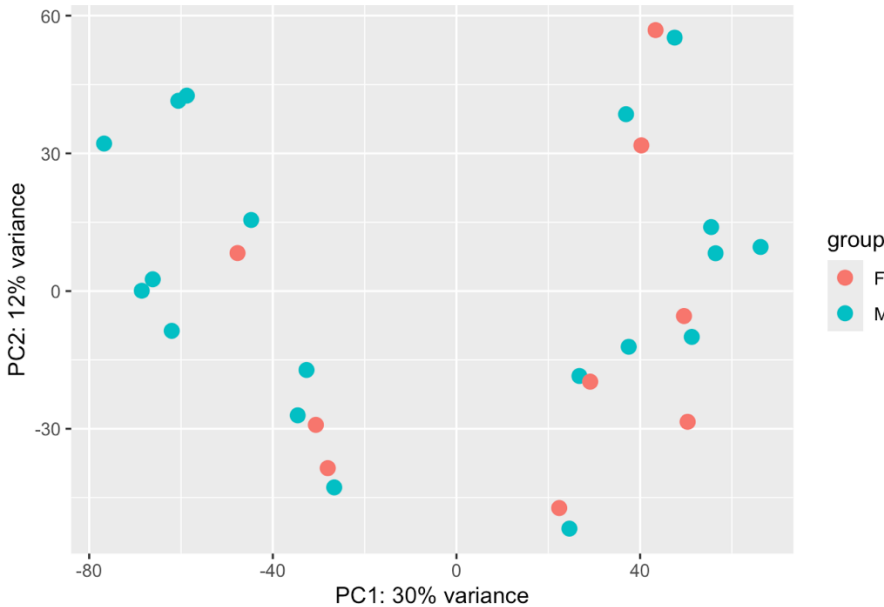


- Day 18 samples form the clearest cluster
- This could be due to differential gene expression when the organism approaches the pupa stage

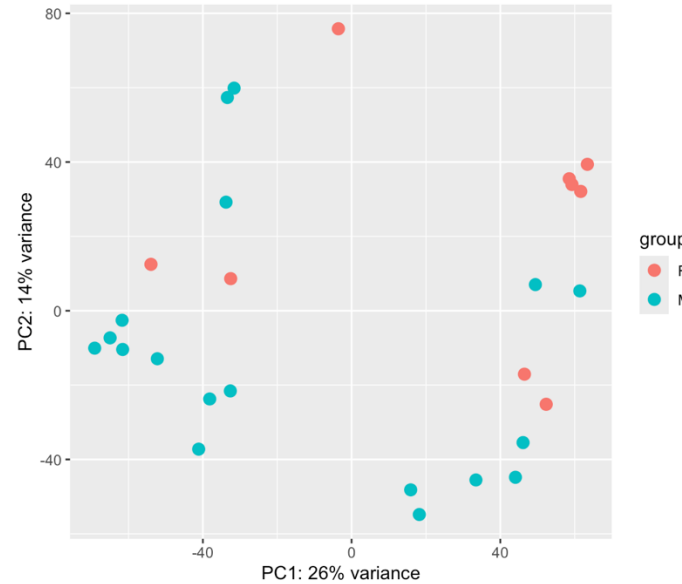
Principal Component Analysis

Differential expression analysis

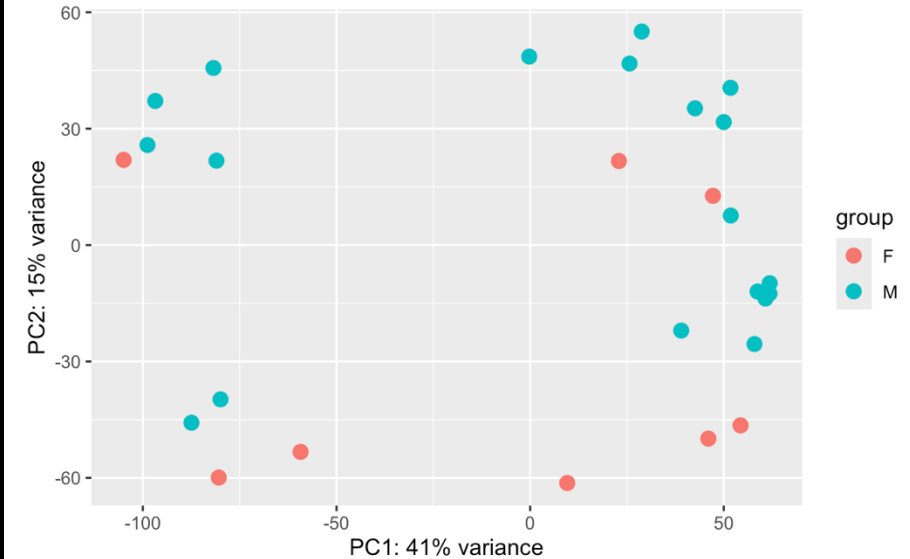
PCA of samples at 14 days



PCA of samples at 16 days



PCA of samples at 18 days

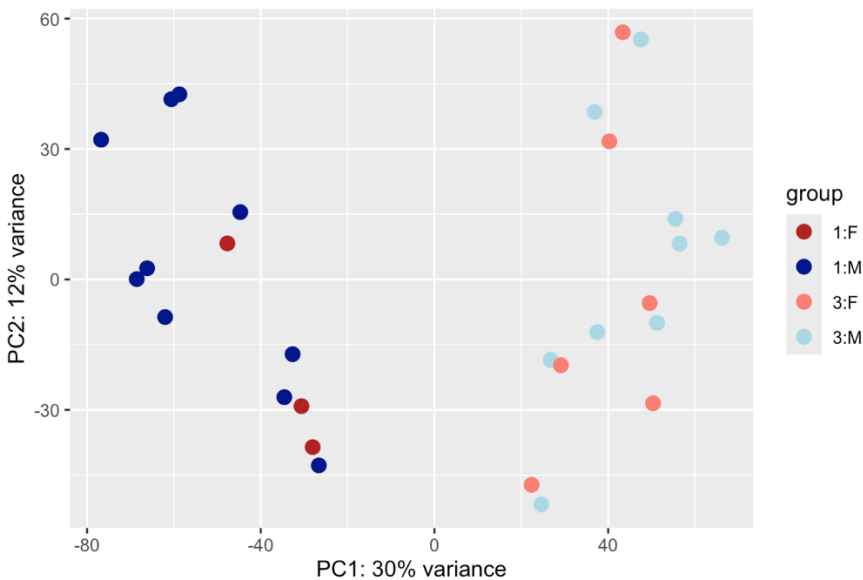


- In day 16 we notice a slight clustering of males, mostly determined by PC2
- Probably more female samples would help identify a better pattern in days 16 and 18

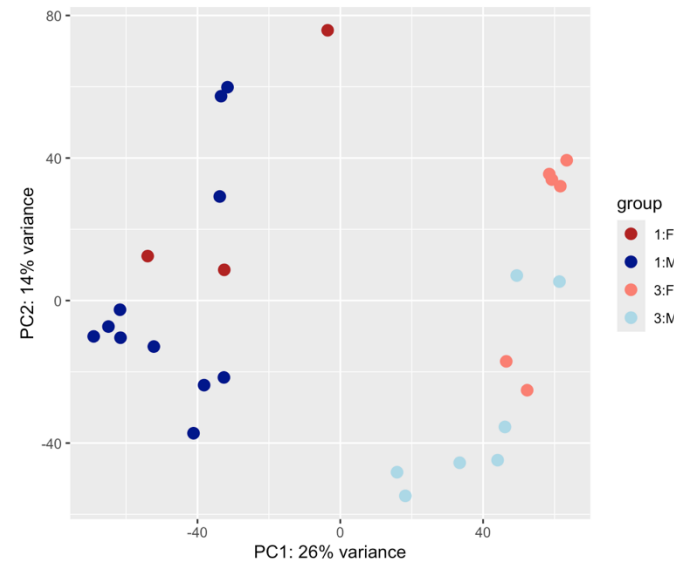
Principal Component Analysis

Differential expression analysis

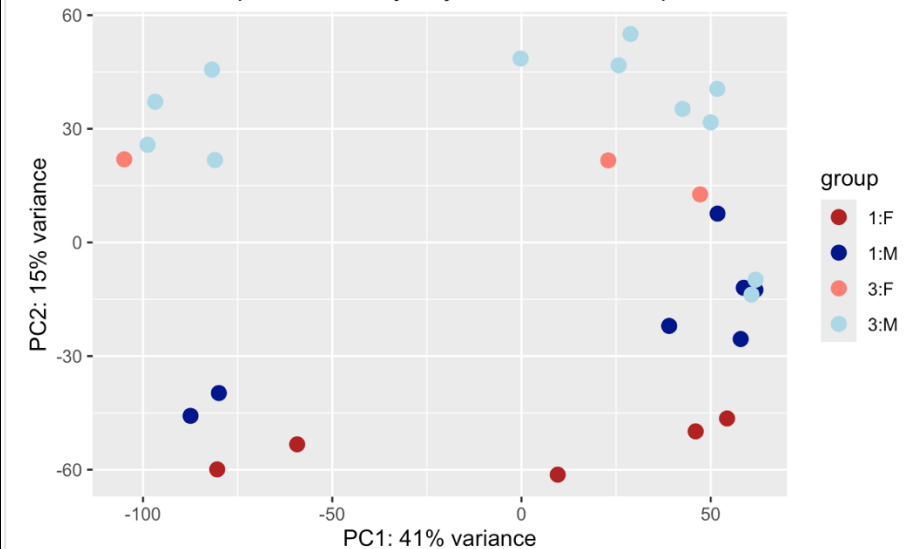
PCA of samples at 14 days by sex and TOR copies



PCA of samples at 16 days by sex and TOR copies



PCA of samples at 18 days by sex and TOR copies



- When sorting the samples by lineages, better patterns become more noticeable, especially in the earliest developmental stage
- In day 14, we notice the two female lineages are closer together than the two male lineages. This could indicate the genetic expression is quite different in males at this developmental stage, possibly explaining the different phenotypes – smaller and bigger males.

Conclusion

- The *Cmac* samples were used to create a transcriptome which is used to assess differential expression in the two lineages
- The sexes do not show large differences caught by the main PCs
- Different gene expression levels were identified between the two lineages, but not necessarily linked to the difference between the TOR copies.
- Day 18 samples show quite different expression patterns compared to day 14 and 16
- Day 14 samples show that the females in both lineages have more similar expression patterns than the males in the two lineages.



Thank you
for listening!!

